NEXUS-IC: Final Complete Solution Document (Unified Edition)

Integrated Institution Grievance Monitoring and Resolution System | AICTE | Problem Statement 26264

Edition 24 October 2026. Merges the 23 October specification (64 features, 10 enhancements, 9 corrections, 74 acceptance tests) with the later architecture visuals (the 6-stage journey and the 9-layer solution architecture). Status: proposed design on synthetic data. It is not a deployed AICTE system, and nothing here claims live AICTE, CPGRAMS, ABC/APAAR or SMS access.

1. Read this first

What this edition does. The earlier PDF defined an internal, officer-facing coordination layer (institution 360, overlap review, NEXUS PREVENT, NEXUS SYNC, SLA, audit). The later visuals added the front half of the journey (verified reporters, the institution registry with ICID, single intake, intelligent routing) and the back half (NEXUS TRACE, PROOF, RESCUE, appeal, IMPACT, RULEBOOK, zero-trust security). This document puts both into one specification and records every place where they disagreed (Section 17).

Conventions.

- Feature IDs F01-F64 and enhancements E01-E10 are kept unchanged from the PDF.
- New feature IDs from the visuals are **N01-N35**. New security controls are **S01-S30**. New acceptance tests are **U01-U32**. The old T01-T30, X01-X22 and D01-D26 tests stay in force.
- P0 is essential and demonstrated, P1 is a competitive differentiator, P2 is planned or optional. Every item is proposed until a test run proves it.
- Anything that needs an official AICTE system, an identity provider (ABC/APAAR, SSO), a messaging gateway or a legal timeline is **contract-gated**: it runs against a synthetic simulator until written approval exists.

2. Inclusion ledger: what came from where

Element	Earlier PDF (23 Oct)	Later visuals	Where in this document
AICTE Permanent Institute ID (PID)	Yes (F01)	Yes	Sections 3, 6

Element	Earlier PDF (23 Oct)	Later visuals	Where in this document
ICID / IIC ID	**No**	**Yes** (both visuals)	**Section 3**, then 6, 12, 13
Student, faculty, other-user verification	Partial (demo accounts, F45)	Yes	Section 5
Pseudonymous reporter ID, consent, no-proxy	No	Yes	Section 5.3
Institution master registry with auto-fetch	Partial (F01-F04)	Yes	Section 6.1
Six-region classification	Region field only	Yes	Section 6.3
Institution 360	Yes (F04)	Extended	Section 6.4
Complaint submission portal	No (internal only)	Yes	Section 7.1
Single Master Case ID	Partial (local ID + source key)	Yes	Section 7.2
Multi-channel intake	Partial (CSV/API adapters)	Yes	Section 7.3
NEXUS PREVENT Stage 1 (intake duplicates)	No	Yes	Section 7.4
NEXUS PREVENT Stage 2 (action conflicts)	Yes (E01, F23-F24)	Yes	Section 8.5
NEXUS ROUTE (intelligent routing)	Partial (assignment, F10)	Yes	Section 8.1
Risk and priority analysis	Partial (demo SLA)	Yes	Section 8.2
NEXUS TRACE (transfer handover)	Yes (F12, 4.7)	Extended	Section 9.2
NEXUS SYNC 2.0	Yes (E02)	Extended	Section 9.3
NEXUS PROOF (closure)	Partial (F16, F26)	Yes	Section 10.1
NEXUS RESCUE (stalled cases)	Partial (E05, E07)	Yes	Section 10.2
Appeal and review	Partial (reopen, F16)	Yes	Section 10.3
NEXUS IMPACT	Yes (E09)	Extended	Section 10.4

Element	Earlier PDF (23 Oct)	Later visuals	Where in this document
NEXUS RULEBOOK	Partial (F39)	Yes	Section 11
True and transparent status	Partial (F09, F11)	Yes	Section 9.1
Competent Authority and governance	Yes (G04)	Yes	Section 12.2
Zero-trust security framework	Yes (F45-F54)	Yes	Section 13
Technology architecture, mobile, adapters	Web stack only	Yes	Section 14
Tests, roadmap, demo	Yes	not shown	Sections 16, 18

3. Where the ICID is included

Assumption to confirm. The PDF never mentions ICID. In the visuals, the registry panel lists "IIC ID (if available)", Institution 360 lists "IIC ID mapping", and the architecture lists "AICTE Permanent Institute ID (PID) / ICID". This edition treats ICID as a **secondary, optional institution identifier** (the IIC ID) that maps to a PID. If ICID means something else at AICTE, only the field definition changes; the touchpoints below stay the same.

Hard rules.

1. The PID is the only authoritative institution key. ICID is a lookup and corroboration identifier.
2. ICID is never used alone to link or merge institutions.
3. If a PID and an ICID disagree, the record goes to the identity-review queue instead of being auto-resolved.
4. A missing ICID is shown as "not available", never as a failed verification.

Every point where ICID appears.

#	Touchpoint	How ICID is included
1	Registry lookup (stage 2)	Enter PID; the fetched profile shows ICID when available
2	Auto-fetch panel	ICID listed with name, state/district, region, type, campus, approval status
3	Institution 360	"IIC ID mapping" block: ICID, mapped PID, source, last verified date

#	Touchpoint	How ICID is included
4	Global search (R32)	ICID accepted as a search key; it resolves to the PID, then PID rules apply
5	Case creation	Case stores a snapshot of PID plus ICID at submission time
6	Matching Stage A	ICID agreement is supporting identity evidence; ICID conflict forces IDENTITY_REVIEW
7	Data model	New collection institution_identifiers (Section 15); institutions gains an icid field
8	API	GET /api/v1/institutions?icid=...; adapter contract carries institutionIdentifiers {pid, icid}
9	Imports	Rows with PID/ICID mismatch are quarantined to data-health review
10	Reports and export	Institution-wise report has an ICID column when permitted
11	Audit	Any ICID add, change or mapping correction is an attributed event
12	Tests	U01-U04 (Section 16)

4. Product purpose and boundary (reconciled)

Purpose. One institution, one view, one complaint journey, accountable resolution. The unified product has two faces on one backend: a complainant-facing intake and tracking portal, and an officer-facing coordination workspace.

Design goals (not measured results). Right institution, verified reporter, right bureau, no lost transfers, faster accountable resolution, transparent governance. These come from the visual's footer and are verified only by the test suites, never asserted as achieved savings.

The product is	The product is not
A verified-reporter intake portal plus institution-wise coordination layer	A replacement for CPGRAMS or a promise of CPGRAMS write access
Explainable routing and overlap suggestions that humans confirm	An automatic judge of guilt, legality or case outcome

The product is	The product is not
A workflow with acknowledgements, rescue alerts and audit trail	An unapproved disclosure channel between bureaus
A configurable SLA and policy engine (RuleBook)	A system with one assumed AICTE-wide 21-day rule
A synthetic-data demonstrator with reproducible tests	A deployed AICTE system or proof of real savings

The six-stage journey.

Stage	Name	Core components	Section
1	Verified users	Student, faculty, other authorized users; privacy-protected identity	5
2	Institution identification and regional classification	Master registry, PID/ICID, six regions, Institution 360	3, 6
3	Grievance submission and case creation	Portal, Master Case ID, multi-channel intake, PREVENT Stage 1	7
4	Intelligent routing and analysis	NEXUS ROUTE, risk and priority, PREVENT Stage 2	8
5	Coordinated processing across bureaus	Workflow, NEXUS TRACE, NEXUS SYNC 2.0	9
6	Resolution and impact	PROOF, RESCUE, appeal, IMPACT	10

5. Stakeholders, verified users and privacy-protected identity (stage 1)

5.1 Stakeholders and entry points

Students, faculty and staff, institutions and colleges, regional offices, AICTE bureaus and cells, and the Competent Authority all use one unified coordination system, each through a role-scoped entry point.

5.2 Verification by user type

User type	Verification evidence	Controls	Status
Student (N01)	ABC / APAAR-linked identity; current enrolment check against institution records	OTP and MFA; no proxy registration	Simulator now; real link contract-gated
Faculty (N02)	Institution staff registry; current employment verification; institutional login / SSO	OTP and MFA; no proxy registration	Simulator now; SSO contract-gated
Other authorized users (N03)	Institution admin records; regional and bureau officials; AICTE staff via SSO	Role-based access control	Demo accounts now
Institution official	Authorized-representative record	Consent-based access, SSO	P1

OTP/MFA is N04 (P0). In the prototype, OTP goes to a local sink, never a real phone.

5.3 Privacy-protected identity

- **Minimal data (N06).** Collect only what verification and routing need.
- **Consent-based verification (N07).** Each verification step records a consent entry (purpose, scope, time, withdrawal). Consent state is shown to the user.
- **Pseudonymous reporter ID (N06).** Officers and other bureaus see a reporter ID such as RP-7K2Q, not a name. The real identity sits in a separate **identity vault** collection.
- **Reveal on need.** Identity is visible only to authorized officers when required. Every reveal needs a purpose, approver and expiry, and is audited. Institution admins and counterpart bureaus never receive reporter identity by default.
- **Alternative verification (N08).** Users without ABC/APAAR or SSO can use an officer-assisted path with manual verification, logged and rate-limited.

5.4 Anti-proxy and fraud signals (N05)

One verified person per account; repeated registration attempts, mismatched enrolment or employment, burst submissions, and shared devices raise a fraud signal that routes to manual verification. A signal never auto-rejects a grievance; a human decides.

6. Institution identification, regions and Institution 360 (stage 2)

6.1 AICTE Institution Master Registry (N10)

The user enters the PID (example format shown in the UI: 1-1234567890). The registry returns a read-only profile:

Fetched field	Notes
College name	Verified aliases shown separately (F03)
State / district	Feeds region classification
AICTE region	One of six (6.3)
Institution type	From registry
Campus details (if any)	Campus hierarchy rules (F02)
IIC ID / ICID (if available)	Section 3
Approval status	Shown with as-of date

If the registry is unavailable, the screen shows UNKNOWN with the reason, never a blank profile or zeros (F48, F56).

6.2 Identity rules

PID is exact-match only and never fuzzy-linked. Similar names with different PIDs are a hard identity block (D06). Ambiguous campuses go to review (F02, T02).

6.3 Region-wise classification (N11)

Cases carry one of six AICTE regions: Northern, Western, Southern, Central, Eastern, North Eastern. The state-to-region mapping lives in a versioned reference table that must be loaded from AICTE's official list, not typed from memory. Reclassification is a reviewer action with a reason. Regional officers see only their delegated region (R50).

6.4 Institution 360 (N12, extends F04)

Contents: complete institution profile; all related grievances, ongoing and past; actions taken by each bureau; region and jurisdiction details; IIC ID mapping; and attention indicators. Visibility follows the single-window contract in the PDF: other bureaus' entries appear as opaque metadata, restricted entries are disclosed as "restricted entries", and completeness is never faked.

Naming change. The visual says "risk indicators". To stay consistent with E07 and the no-guilt-score rule, this edition calls them **attention indicators**: transparent operational reasons such as overdue tasks, stalled acknowledgements and source conflicts. They are not an institutional wrongdoing or compliance score.

7. Grievance submission and case creation (stage 3)

7.1 Grievance submission portal (N13)

- Structured form with category and sub-category.
- Guided fields with examples for each field.
- Upload of supporting documents. Allowed types and size are enforced, files are malware-scanned and hashed (S12), and access is scoped.
- Institution is linked automatically from the verified PID.
- Preview and confirm before submit; the user can edit until confirmation.

7.2 Case creation (N14)

- **Single Master Case ID** for the whole journey, for example NX-2026-000123 (format illustrative). The same ID survives routing, transfers and appeals.
- **Original external references are preserved.** An imported CPGRAMS or bureau reference stays as externalRefs[] next to the Master Case ID. The PDF rule stays: the source-native ID is never overwritten and (sourceSystem, sourceCaseRef) remains unique (T09).
- **Duplicate-registration validation** before the case is created.
- **Confirmation** of institution and user details.

7.3 Multi-channel intake (N15)

Channel	Status	Note
Web portal	P0	Responsive React app
Mobile app	P1 as installable PWA; native Android/iOS is P2	Same API and same verification
Email	P1	Inbound to a monitored mailbox; sender must complete verification before a case is created
API (future)	P2, contract-gated	For approved partner systems

Channel	Status	Note
Import of existing systems	P1	CSV preview, dry-run, idempotent commit (F15, F49)

7.4 NEXUS PREVENT Stage 1: duplicate complaint check at intake (N16)

Before submission, the system compares the draft with open and past cases for the same verified institution.

1. Detect similar or duplicate complaints (same institution, same issue concept, overlapping period).
2. Check the same institution plus same issue.
3. Alert the user before submitting.
4. Allow a new case if the complaint is genuinely different.
5. Explain why it looks similar.

Privacy guard (added). The hint shown to a complainant refers only to that complainant's own earlier cases, or to a generic "a similar open matter exists" notice. It never shows another person's text, identity or case number. A match is advisory, is logged with the user's choice, and never blocks or merges anything.

Continued in the next sections: routing, coordinated processing, resolution, RuleBook, security, architecture, data model, API, features N01-N35, tests, roadmap.

8. Intelligent routing and analysis (stage 4)

8.1 NEXUS ROUTE: explainable jurisdiction engine (N18)

Routing uses several factors together: grievance category and sub-category; institution type and AICTE region; bureau and cell jurisdiction rules; existing ongoing cases for the same institution; policy and regulatory guidelines; and a confidence score with its explanation.

Routing decision (always explainable).

- Primary bureau or cell.
- Secondary bureau or cell, if required.
- Manual review, if the case is ambiguous or confidence is below the RuleBook threshold.
- Reason for routing, stored with the rule version.

Bureaus and cells in the architecture include the Regulation Bureau (RB), the Public Grievance Redressal Cell (PGRC), Regional Offices, the Academic Bureau, the Approval Process bureau, and other cells as configured.

Guardrails. Deterministic RuleBook rules run first; any ML ranking is optional and cannot override them. An officer can override a route only with a recorded reason. Routing assigns work; it does not decide the merits of a complaint and does not move ownership later (later moves go through NEXUS TRACE). Demo rules are labelled DEMO-ROUTE-v1 and are synthetic.

8.2 Risk and priority analysis (N19)

- Automatic priority: normal, high or urgent, from category, sensitivity and age.
- Sensitivity detection (for example ragging or safety) from category plus versioned keyword rules. A human confirms every sensitive flag, and sensitive cases get restricted handling.
- SLA timeline set by category from the RuleBook, with a policy clock (Section 11).
- Policy and compliance flags for officers.
- The priority reason is shown to the officer. It is a workload-prioritization signal, never a guilt or institutional compliance score. Urgent maps to the synthetic DEMO-CASE-PRIORITY-v1 policy in the demo.

8.3 Overlap detection (unchanged from the PDF)

The seven-signal weighted matcher (issue 25, time window 15, category 10, document or subject 15, remedy 10, shared entities 10, semantic 15), the identity gate and the thresholds all stay as specified, including the worked 93.75 example and the exact banner text **Potential Overlapping Cases – Coordination Required** on /dashboard, /institutions/:id and /review. The matcher now also receives ICID as supporting identity evidence (Section 3).

8.4 Routing output states

REGISTERED -> ROUTED (with decision record) -> ASSIGNED (officer named). Ambiguous cases wait in a human triage queue; they are never silently assigned.

8.5 NEXUS PREVENT Stage 2: duplicate action and conflict detection (N17)

This is the E01 pre-action checker, now shown as stage 2 of PREVENT. It detects overlapping actions across bureaus, repeated document requests, and conflicting instructions; shows related ongoing cases; suggests coordination before proceeding; and

raises real-time alerts to officers. Output is CLEAR, REVIEW or UNKNOWN with reasons and counter-reasons, using rules C1-C5 from the PDF. Missing or stale feeds return UNKNOWN. It never blocks an official act.

9. Coordinated processing across bureaus (stage 5)

9.1 Truthful case lifecycle and status (N28)

Statuses: Registered (also called Submitted), Assigned, Under Process, Transfer Requested, Transfer Accepted, Under Review, Resolution Proposed, Verified/Approved, Closed.

Four parallel tracks per case: case status, transfer status, current owner, and SLA clock with escalation. A transferred case must never revert to Registered or Submitted.

Lifecycle status	Mapped analytic status (F09)	Note
Registered	PENDING	Native status kept alongside
Assigned	PENDING	Owner named
Under Process, Under Review	IN_PROCESS	
Transfer Requested, Transfer Accepted	IN_PROCESS plus transferStatus	Counted in the transferred metric by event
Resolution Proposed, Verified/Approved	IN_PROCESS	Awaiting closure gate
Closed	DISPOSED	Escalated is a flag, not a status

What the complainant sees (true and transparent status). The exact current status; no false closure; the accountable owner shown as office and role; real-time updates; a complete status history; the SLA clock and escalation alerts in summary form. Internal notes, other bureaus' narratives and reporter identity stay hidden.

9.2 NEXUS TRACE: zero-loss handover (N20)

Sequence: Transfer Requested -> Acceptance Pending -> Transfer Accepted.

1. The original Case ID remains the same.
2. Every transfer must be acknowledged by the receiving bureau.

3. Ownership is always visible: at any moment a case has one accountable owner, or an open transfer with a named sender who stays accountable.
4. Escalation fires if not acknowledged in time (synthetic DEMO-TRANSFER-ACK-v1: 2 business days).
5. Full transfer history with timestamps.
6. A rejection returns to the sender's queue with its reason; nothing is reassigned silently.

"No lost transfers" is a design goal, enforced by invariant tests T15, U20-U22, and not a measured result.

9.3 NEXUS SYNC 2.0 additions (N21)

Builds on E02 and its state machine. Adds: a shared view of all related cases for an institution; task coordination and document sharing (documents move only through scoped, time-bound grants); internal discussions and notes visible to officers in scope only; a common communication log; prevention of conflicting instructions; and escalation of unresolved inter-departmental cases. Task completion is never case disposal.

9.4 Processing workflow

Assigned to bureau -> Under Process -> Transfer (if required) -> Verification and Action -> Resolution Proposed. Each step records actor, time, remark and evidence.

10. Resolution and impact (stage 6)

10.1 NEXUS PROOF: evidence-based closure (N22, N23)

- Resolution needs evidence-based support and mandatory action records.
- Final documents are uploaded and hashed.
- An authorized officer approves; the approver must differ from the proposer.
- The complainant's feedback is collected after resolution is proposed or verified (N23). Feedback is recorded and shown to reviewers; it does not itself close, reopen or change a case.
- **Closure gate.** Closure is a human action, but the system blocks it until every task is complete, every transfer is acknowledged, evidence is attached and approval exists. The system never closes a case on its own.

10.2 NEXUS RESCUE: stalled-case recovery (N24)

Detects stalled or repeatedly transferred cases; alerts supervisors automatically; shows the reason for delay; suggests the next action; and prevents indefinite pendency.

Synthetic triggers (DEMO-RESCUE-v1, configurable in the RuleBook): three or more transfers; no recorded event for 7 calendar days on an open case; a transfer acknowledgement past its escalation point; a coordination task overdue twice. Alerts are deduplicated. Reasons point to work items, not to misconduct, and a late-arriving feed never counts as officer delay (F43).

10.3 Appeal and review (N25)

The complainant can appeal if required. Flow: Closed -> Appeal Filed -> Under Appeal Review (level 1 to n) -> Upheld, Modified or Remanded (reopened). Additional evidence can be submitted; the final decision is tracked. Appeals are new events; the original record is untouched (F16). The appeal reviewer must differ from the original resolver. The appeal window and number of levels come from the RuleBook, not from assumption.

10.4 NEXUS IMPACT: analytics and continuous improvement (N26)

Resolution-time tracking; reduction in duplicate actions; bureau-wise and region-wise insights; policy-level recommendations; downloadable reports. Safeguards from E09 apply: results are labelled synthetic until a pilot exists, every percentage shows numerator, denominator and scope, "action flagged" is never counted as "action avoided", and a policy recommendation must cite the report rows it rests on.

11. NEXUS RULEBOOK: policy configuration (N27)

One versioned registry that drives routing, SLA, transfer and escalation behaviour without code changes.

Rule family	Drives	Demo ID (synthetic)
Jurisdiction rules	Which bureau or cell owns a category in a region	DEMO-ROUTE-v1
Routing rules	Primary and secondary route, confidence threshold	DEMO-ROUTE-v1
SLA timelines by category	Case clock, near-due, overdue	DEMO-CASE-STANDARD-v1, DEMO-CASE-PRIORITY-v1
Transfer policies	Acknowledgement window	DEMO-TRANSFER-ACK-v1
Escalation matrix	Who is told, and when	per policy above
Coordination and task clocks	SYNC and task deadlines	DEMO-SYNC-ACK-v1, DEMO-TASK-v1

Rule family	Drives	Demo ID (synthetic)
Rescue triggers	Stalled-case detection	DEMO-RESCUE-v1
Appeal rules	Window, levels, reviewer separation	not seeded (POLICY_NOT_CONFIGURED)

Controls. Every edit creates a new version with an effective date and is never retroactive. Changes need two-step approval (requester differs from approver) and a dry-run against the fixture set first. Versions are immutable and audit-ready. In production mode demo rules cannot be selected; a missing approved rule gives POLICY_NOT_CONFIGURED or UNKNOWN. No AICTE-wide deadline is assumed.

12. Workspaces, governance and outcomes (layers 8-9)

12.1 Role-based workspaces

Workspace	Content	Route
Institution 360	Profile, PID/ICID, cases, actions, region	/institutions/:id
Officer dashboard	My cases, tasks, SLA, actions	/dashboard
Regional view	Region-wise cases and analytics	/dashboard (regional scope)
Bureau/Cell queue	Assigned cases and coordination	/cases, /sync
Competent Authority workspace	Monitor, intervene, approve	/authority
Notifications	In-app, email, SMS (SMS sandbox until approved)	notification centre
Reports and analytics	Trends, performance, region-wise insights; ICID column when permitted	/reports
Audit trail	Complete history of all actions	/audit
Feedback and appeal	User feedback and appeal handling	/track/:caseId (complainant), /review
NEXUS IMPACT metrics	Outcome and impact measurement	/impact

12.2 Governance and oversight

The Competent Authority dashboard shows all institutions and all cases in one view; pending, overdue and transferred cases; region-wise and bureau-wise insights; approvals and escalations; notifications; audit and evidence history; impact metrics; and feedback. Sensitive approvals need requester and approver to differ. Authority never gains automatic access to reporter identity or private narratives (D16).

12.3 Extended role matrix (additions to the PDF matrix)

Role	Can see	Cannot see
Complainant (student or faculty)	Own cases, own status history, own documents	Other users' cases, internal notes, officer identities beyond office and role
Institution admin	Institution profile, cases about their institution in the redacted institution-facing form, requests addressed to them	Reporter identity, other bureaus' internal notes
All officer roles from the PDF	As specified in Section 10 of the PDF matrix	Unchanged

12.4 Designed outputs and outcomes

Faster routing, clear ownership, no lost transfers, reduced duplication, timely resolution, verified accountability, real-time monitoring, institution-wise and region-wise insights. For managers: monitoring of pending, disposed, transferred and escalated cases; truthful status tracking; cross-bureau overlap detection; exact ownership at every stage; alerts for approaching timelines; complete audit history; decision support. These are design outputs; each is proven only by the tests in Section 16.

13. Security and privacy framework (zero-trust design)

Encryption, certification and compliance below are design proposals; none is claimed as achieved. "Source" shows whether a control came from the earlier PDF, the later visuals, or is added in this edition to close a gap the merge exposed.

ID	Control	Source	Proved by
S01	Zero trust: every request re-authorized server-side	PDF, visual	T14, X22

ID	Control	Source	Proved by
S02	RBAC by bureau, region and case level	Visual	U31
S03	Field-level ABAC on read, write, search, export, socket, file	PDF	T11, T14
S04	Minimal data collection	Both	U05
S05	Consent-based identity verification with consent ledger	Visual	U10
S06	Pseudonymous reporter ID in all officer views and exports	Visual	U08
S07	Identity vault separate from case data; audited, expiring reveal	Added	U09
S08	Encryption at rest and in transit (for real deployment)	Both	Deployment review
S09	Tamper-evident audit log (hash chain, not a blockchain claim)	Both	T30
S10	Secure document storage with authorization on every download	Both	D21
S11	Server-proxied or signed downloads; raw storage keys never exposed	PDF	T13
S12	Upload type and size checks, malware scan, file hash	Added	U11
S13	Regular security audits and compliance reviews	Visual	Audit schedule, S34
S14	OTP and MFA	Visual	U05, U06
S15	Anti-proxy checks, fraud signals, rate limits	Visual, added	U07
S16	HttpOnly Secure session cookies, no tokens in localStorage	PDF	Auth tests
S17	Blind, opaque cross-bureau notices	PDF	T12, X14
S18	Purpose-bound, time-boxed disclosure grants	PDF	T13, X13
S19	Competent Authority cannot bypass field ABAC	PDF	D16
S20	Requester differs from approver on sensitive approvals	PDF	D15, U25

ID	Control	Source	Proved by
S21	Complainant sees own cases only; IDOR denied	Added	U31
S22	Institution admin never sees reporter identity	Added	U32
S23	Intake similarity hints leak no third-party data	Added	U14
S24	Internal notes never reach the complainant	Added	U23
S25	Anti-retaliation: no reporter identity in notifications, exports or logs; sensitive cases get restricted handling	Added	U18, U08
S26	Consent withdrawal and retention hooks (DPDP-informed; legal review needed)	Visual, PDF F52	U10
S27	Idempotent outbox; email and SMS carry no narrative or PII	PDF	D17-D19
S28	No complainant text in logs	PDF	Log review
S29	Backup, restore, import replay	PDF	T09
S30	Threat model and penetration-test checklist before any pilot	Added	Pre-pilot gate

GIGW 3.0 and WCAG 2.1 AA practices guide the UI; "accessible by design" is not a certification.

14. Technology architecture

The architecture stays a modular JavaScript monolith: React JSX with Vite, Express in plain JavaScript on Node.js, MongoDB with Mongoose, Tailwind, React Router, TanStack Query, Recharts, optional React Flow. No TypeScript. No Digital Twin.

Visual layer	Implementation in this specification	Priority
Web portal (responsive UI)	React JSX app with complainant and officer areas	P0
Mobile app (Android / iOS)	Installable PWA on the same API first; native apps later	P1 PWA, P2 native
Database (secure, scalable)	MongoDB replica set, indexed collections	P0

Visual layer	Implementation in this specification	Priority
Document storage (encrypted)	Private object store, encrypted, scanned, hashed	P0
APIs and integration layer	REST API plus adapters for AICTE systems, institution registry, CPGRAMS	Synthetic now; real contract-gated (P2)
Integration adapters	Source adapter contract with snapshots and watermarks (PDF Section 11)	P0 synthetic
Business logic and workflow engine	Rule-based, driven by the RuleBook, configurable without code changes	P0
Analytics and reporting engine	One query builder for UI, CSV and PDF (PDF 4.9)	P0

New server modules beside the PDF's modules: identity (verification, consent, vault), registry (PID/ICID), intake, routing, rescue, appeals, rulebook, feedback. New client pages: Track.jsx (complainant), Submit.jsx, Verify.jsx, Appeals.jsx, RuleBook.jsx.

15. Data model, APIs and feature catalogue

15.1 New and extended collections

Collection	Key fields	Invariant
institution_identifiers	institutionId, pid, icid, source, verifiedAt, status	PID unique; ICID never links alone; conflict goes to review
reporter_accounts	pseudonymousReporterId, userType, verificationMethod, verifiedAt, status	No raw identity stored here
identity_vault	reporterId, encrypted identity payload, keyRef	Reveal only via approved, expiring, audited request
consent_records	reporterId, purpose, scope, grantedAt, withdrawnAt	Append-only

Collection	Key fields	Invariant
cases (extended)	masterCaseId, externalRefs[], channel, subCategory, priority, sensitivityFlags, routingDecisionId, pidSnapshot, icidSnapshot	Master ID stable; (sourceSystem, sourceRef) unique
routing_decisions	caseId, primary, secondary, confidence, reasons, ruleVersion, overriddenBy, reason	Immutable
intake_similarity_checks	draftHash, reporterId, shownHints, userChoice	No third-party text stored
closure_records	caseId, evidenceRefs, proposedBy, approvedBy, gateResults, packetHash	Approver differs from proposer
feedback_records	caseId, reporterId, rating, text, at	Does not change status
appeals	caseId, level, filedAt, evidenceRefs, reviewerId, outcome	Original case events untouched
rescue_alerts	caseId, trigger, ruleVersion, supervisorId, state, dedupeKey	Unique dedupeKey
communication_log	caseId or coordinationId, from, to, channel, at	Append-only
rule_book_versions	family, body, effectiveFrom, approvedBy, requestedBy	Immutable; non-retroactive

15.2 New API routes (all server-authorized; error codes as in the PDF)

Method	Route	Purpose
POST	/api/v1/auth/verify/student, /faculty	Verification (simulator in prototype)
POST	/api/v1/consents, /consents/:id/withdraw	Consent ledger
POST	/api/v1/identity/reveal-requests	Audited reveal request
GET	/api/v1/registry/institutions/:pid	Registry lookup and auto-fetch, includes icid

Method	Route	Purpose
GET	/api/v1/institutions?icid=	ICID search, resolves to PID
POST	/api/v1/grievances/draft, /check-similar	Draft and intake similarity check
POST	/api/v1/grievances	Submit; returns masterCaseId
POST	/api/v1/routing/:caseId/decide, /override	Route and override with reason
GET	/api/v1/track/:caseId	Complainant status view (own cases only)
POST	/api/v1/cases/:id/closure/propose, /approve	PROOF closure gate
POST	/api/v1/cases/:id/feedback	Complainant feedback
POST	/api/v1/cases/:id/appeals, /appeals/:id/decide	Appeal and review
GET	/api/v1/rescue/alerts	Stalled-case alerts for supervisors
GET, POST	/api/v1/rulebook/families, /versions	RuleBook read and versioned change

15.3 Feature catalogue N01-N35 (added to F01-F64)

ID	Feature	Pri	Section
N01	Student verification (ABC/APAAR-linked, enrolment check)	P0 simulated	5.2
N02	Faculty verification (staff registry, employment, SSO)	P0 simulated	5.2
N03	Other authorized user onboarding	P0	5.2
N04	OTP and MFA	P0	5.2
N05	Anti-proxy and fraud signals	P1	5.4
N06	Pseudonymous reporter ID and identity vault	P0	5.3
N07	Consent-based verification and ledger	P0	5.3
N08	Alternative verification path	P1	5.3
N09	ICID / IIC ID field, mapping and search	P0	3
N10	Institution master registry lookup and auto-fetch	P0	6.1

ID	Feature	Pri	Section
N11	Six-region classification	P0	6.3
N12	Institution 360 extended with attention indicators	P0	6.4
N13	Grievance submission portal	P0	7.1
N14	Master Case ID, external refs, duplicate-registration check	P0	7.2
N15	Multi-channel intake	P0 web, P1 others	7.3
N16	PREVENT Stage 1: intake similarity check	P1	7.4
N17	PREVENT Stage 2: action conflict detection	P1	8.5
N18	NEXUS ROUTE	P0	8.1
N19	Risk and priority analysis	P0	8.2
N20	NEXUS TRACE extended	P0	9.2
N21	SYNC 2.0 notes, communication log, conflict prevention	P1	9.3
N22	NEXUS PROOF closure gate	P0	10.1
N23	Complainant feedback	P1	10.1
N24	NEXUS RESCUE	P1	10.2
N25	Appeal and review	P1	10.3
N26	NEXUS IMPACT extended, downloadable reports	P1	10.4
N27	NEXUS RULEBOOK	P0	11
N28	True and transparent complainant status	P0	9.1
N29	Complainant workspace	P0	12.3
N30	Institution admin scoped view	P1	12.3
N31	Encrypted document storage with malware scan	P0	13
N32	Integration adapters for real AICTE and CPGRAMS systems	P2	14
N33	Native Android and iOS apps	P2	14
N34	Recurring security audit programme	P1	13

ID	Feature	Pri	Section
N35	Minimal status notices by email and SMS to complainants	P1	12.1

16. Acceptance tests added: U01-U32

The T01-T30, X01-X22 and D01-D26 tests remain in force. A test counts as passed only with an archived real run.

ID	Verifiable behaviour
U01	PID lookup returns the profile, including ICID when available
U02	ICID search resolves to the PID; ICID alone never links institutions
U03	PID and ICID disagreement goes to identity review, not auto-resolved
U04	Missing ICID shows "not available"; case creation still works
U05	Student without valid enrolment cannot submit a verified grievance
U06	Faculty with inactive employment is rejected
U07	Proxy registration attempt is blocked and flagged
U08	Reporter ID is pseudonymous in officer views and exports
U09	Identity reveal needs purpose and approver, expires, and is audited
U10	Consent withdrawal is handled per policy and recorded
U11	Form validation, preview, and edit-before-confirm work; bad uploads are rejected
U12	Master Case ID is unchanged after routing, transfer and appeal
U13	External reference is preserved; duplicate registration is blocked

ID	Verifiable behaviour
U14	Similarity alert explains why, lets the user proceed, shows no third-party data
U15	Email from an unverified sender creates no case
U16	Routing returns primary, secondary and reason; ambiguous goes to manual review
U17	Route override requires a reason and is audited
U18	Sensitive category is flagged, human-confirmed, and restricted
U19	Priority change alters the clock via a RuleBook version, never retroactively
U20	Transfer needs acknowledgement, escalates after the window, owner always visible
U21	Rejected transfer returns to the sender's queue with its reason
U22	A transferred case never reverts to Registered
U23	Internal notes are invisible to the complainant
U24	Closure is blocked until the gate passes
U25	Closure approver differs from proposer
U26	Feedback is stored and does not change status
U27	RESCUE fires on three transfers or a stall, once per case (deduplicated)
U28	Appeal adds events, leaves the original intact, uses a different reviewer
U29	RuleBook edit makes a new version with dry-run and two-step approval
U30	IMPACT report downloads with denominator and synthetic label; recommendations cite rows
U31	Complainant sees only own cases; IDOR attempts are denied
U32	Institution admin sees no reporter identity

Total acceptance suite: T01-T30, X01-X22, D01-D26 and U01-U32 (110 checks).

17. Where the two sources disagreed, and the resolution

Topic	Earlier PDF	Later visuals	Resolution
Audience	Internal officer tool; not an intake system	Complainant-facing portal	Both faces exist on one backend; complainant role is strictly scoped (Section 12.3)
Risk indicators	No institutional guilt or compliance score	"Risk indicators" on Institution 360	Renamed attention indicators with visible reasons
Auto-routing	No automatic decisions	Automatic jurisdiction engine	Routing assigns work only; ambiguous goes to a human; override with reason
Closure	No automatic closure	"Closure only after all actions completed"	System gates closure; a human performs it
Real-time status for users	Minimal disclosure	Full transparent status	Complainant sees own status only; internal notes hidden
Duplicate check at intake	PREVENT is a pre-action officer check	AI case matching at intake	Added as Stage 1 with a privacy guard
Mobile app	JavaScript web stack only	Android / iOS	PWA first; native is P2
ICID	Absent	Present	Secondary identifier; PID stays authoritative
SLA values	Seeded demo only	Category-wise SLA	RuleBook with demo defaults; production without approved rule shows POLICY_NOT_CONFIGURED
Outcome claims	No claimed savings	"No lost transfers", "faster resolution"	Treated as design goals proven by tests, not as results

18. Roadmap, decisions needed and definition of done

Staged delivery (extends PDF Section 20).

Stage	Adds	Evidence
S0 Foundation	Repo, seed, auth, demo and complainant accounts	Reproducible install and reset
S1 Mandatory PS plus front half	F01-F13, F29-F35, N01-N04, N06, N07, N09-N14, N31	R01-R55 demonstrable; U01-U13
S2 Trust and routing	F14-F19, F48-F49, N18, N19, N27, N28	Hard negatives, U16-U19, U29
S3 SYNC and TRACE	F20-F22, E02, E06, N20, N21	Handshake, receipt, U20-U23
S4 PREVENT	F23-F24, E01, N16, N17	U14, X01-X04
S5 Closure and recovery	N22-N26, N29, N30, N35	U24-U32
S6 Differentiators	F05-F06, F28, F36-F38, E03-E10	Graph, radar, impact
S7 Release evidence	T, X, D and U suites, screenshots, limitations	No critical privacy or identity failure

Decisions and approvals needed from AICTE or the team.

1. Confirm what ICID stands for and its authoritative source (Section 3).
2. Approve the complainant-facing scope, or keep the portal as a prototype module only.
3. Obtain written access terms for ABC/APAAR, institution registries, SSO, CPGRAMS and SMS gateways; until then use simulators.
4. Provide the official state-to-region mapping, category list, jurisdiction rules, SLA timelines, appeal levels and escalation matrix for the RuleBook.
5. Legal and security review of consent, pseudonymous identity, retention and reveal rules before any pilot.

Definition of done. Every requirement R01-R55 is mapped and working or listed as a gap; every P0 feature passes its UI and API tests; the two-bureau judge scenario runs without database edits; no restricted data leaks in any negative-role test; missing feeds never display as zero; and every figure comes from an actual local run. Not claimed: live AICTE data, government approval, measured savings, or certified compliance.

References

- [1] AICTE Public Grievance Redressal Cell: https://www.aicte-india.org/bureaus/grievance-redressal

- [2] AICTE Centralized Support System: https://www.aicte.gov.in/centralized-support-system
- [3] CPGRAMS public portal, DARPG: https://pgportal.gov.in/
- [4] Guidelines for Indian Government Websites and Apps (GIGW 3.0): https://guidelines.india.gov.in/guidelines/
- [5] DARPG comprehensive public grievance guidelines, 23 August 2024: https://www.darpg.gov.in/sites/default/files/Comprehensive_guidelines_for_handling_the_Public_Grievances.pdf
- [6] MeitY Digital Personal Data Protection Rules, 2025 (check applicable enforcement stage with counsel)
- Internal: NEXUS-IC Final Complete Solution PDF (23 October 2026); NEXUS-IC architecture visuals (6-stage journey and 9-layer solution architecture).